What Are Phases? Geographic Structure, Neutral Copying, and the Search for Social Boundaries in the Lower Mississippi Valley

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Abstract

Culture historians built phases to describe continuous spatial and temporal variation in artifact classes, yet phases are interpreted as social units and treated as communities. We ask why phases look like groups of people, and whether the St. Francis basin phases of the Lower Mississippi Valley are more than geographic clumps. On 28 time-averaged decorated-ceramic assemblages, clustering on site coordinates alone recovers Phillips's phases for all but two, a spatial field model places 93 percent of between-site compositional variation on a smooth geographic surface, and the phase boundaries run through the least-occupied ground between settlement clusters. The difference between the basin's two spatial clusters falls within what neutral copying at matched diversity produces. Three common decorated classes vary between sites far more than neutral copying produces, with or without regional drift, while rare classes do not. Rim attributes recorded independently on 19 sites are structured differently from the decoration. Two models built to detect bounded units fail to recover planted units, because a gradient sampled at settlement gaps and a step there leave the same record. The phases are real as a geographic pattern, unresolvable as bounded units on this sample, and not demonstrated as social units.

Keywords: cultural transmission, neutral models, isolation by distance, phases, Mississippian archaeology, Parkin phase

Resumen

Los historiadores culturales construyeron las fases para describir la variación espacial y temporal continua de las clases de artefactos, pero las fases se interpretan como unidades sociales y se tratan como comunidades. Preguntamos por qué las fases parecen grupos de personas y si las fases de la cuenca del río St. Francis, en el valle inferior del Misisipi, son algo más que agrupaciones geográficas. En 28 conjuntos de cerámica decorada promediados en el tiempo, el agrupamiento basado solo en las coordenadas de los sitios recupera las fases de Phillips para todos salvo dos, un modelo de campo espacial sitúa el 93 por ciento de la variación composicional entre sitios en una superficie geográfica suave, y los límites de las fases atraviesan el terreno menos ocupado entre las concentraciones de asentamientos. La diferencia entre las dos agrupaciones espaciales de la cuenca queda dentro de lo que produce la copia neutral con diversidad equiparada. Tres clases decoradas comunes varían entre sitios mucho más de lo que produce la copia neutral, con o sin deriva regional, mientras que las clases raras no lo hacen. Los atributos de los bordes, registrados de forma independiente en 19 sitios, se

estructuran de manera distinta a la decoración. Dos modelos construidos para detectar unidades delimitadas no recuperan las unidades impuestas, porque un gradiente muestreado en los vacíos de asentamiento y un escalón en ellos dejan el mismo registro. Las fases son reales como patrón geográfico, no resolubles como unidades delimitadas en esta muestra y no demostradas como unidades sociales.

Palabras clave: transmisión cultural, modelos neutrales, aislamiento por distancia, fases, arqueología misisipiana, fase Parkin

Introduction

Phases are the units in which much of the North American archaeological record is organized. Willey and Phillips (1958:22) defined the phase as "the practicable and intelligible unit of archaeological study": a set of traits distinctive enough to separate it from other such units, confined to a locality or region and a brief interval of time. Nothing in that definition is social. Pottery types and the other classes that define a phase vary continuously across space and through time, and the people who built phases knew they were dividing that variability into units for description. However, the units are often interpreted as social groups. Willey and Phillips (1958:48–50) set out the chain of reasoning that equates component with community and phase with society, conceded that "in practice it does not work," and then proposed that archaeologists "may act as if" phases had social reality. In the Lower Mississippi River Valley, the survey phases were built "to elucidate sociopolitical boundaries" (Haley 2014:41), and Williams, defining the southeast-Missouri phases, wrote that "in most cases, phase probably equals 'society' " (Williams 1954:25; quoted in Fox 1998:33). A descriptive unit became a people.

That step carries a premise that is rarely stated. It holds that assemblages more similar to one another than to their neighbors record a bounded set of people who interacted mainly with each other. On that premise, a phase defined by a single class of artifacts becomes a hypothesis about social interaction. It remains a hypothesis until independent evidence from other artifact classes, ancient DNA, linguistics, or documented history bears on it. Pairing a phase with a dominant center site adds to what the premise must support. A grouping becomes an inferred polity before independent evidence for its social boundaries has been evaluated (Fox 1998), following the expectation that a ranked society leaves a hierarchy of settlement types with its center at the top (Peebles and Kus 1977:431).

Why, then, do phases look like groups of people? When culture historians examined the record, they saw clumps of deposits whose assemblages resembled one another more than they resembled assemblages elsewhere. That observation matched their expectations, and the phases they drew documented those patterns. Earlier critiques have shown that the clumps are not sharply bounded. Fox (1998) used cluster analysis and ordination to show that the southeast-Missouri phases fail as both groups and classes, with member assemblages falling into

statistical clusters at random rather than by phase. Mainfort (2003a:180) found "little evidence for the integrity of the traditional late period phases" of the Lower Mississippi Valley, even though "most sites assigned to most phases are located in the general vicinity of each other in ordination space." Lipo (2001:60) suggested through frequency seriation that "the phases themselves might be simply arbitrary spatial and temporal slices of a single interacting population that is much larger in size." These studies show that the phases do not separate completely. They leave open why the phases appeared to form coherent groups in the first place, and what the clumps are.

Similarity alone cannot answer that question, for structural rather than statistical reasons. Phillips (1970) drew the phases tested here by sorting assemblages on their proportions of decorated pottery types and on their proximity. A scheme recovered from the same frequencies and the same map records how the scheme was drawn, not how its units behaved (Willey and Phillips 1958:27–28). Ordination arranges assemblages by resemblance and cannot distinguish a bounded learning community from copying that declines gradually with distance, because both produce neighbors that resemble each other.

The phases of the Lower Mississippi Valley offer the classic case. Its late precontact phases are well defined, long established, and in active use. Regional phases include Kent (House 1991), Nodena (Dan F. Morse 1990), Parchman (Starr 1984), Tipton, Walls, and Jones Bayou (Smith 1990), and Parkin (Phyllis A. Morse 1990). Figure 1 shows four of them, Nodena, Parkin, Walls, and Kent, as Phillips (1970) drew them. The Parkin phase stands out because it contains the Parkin site, the largest moated and palisaded town in the St. Francis basin of northeastern Arkansas. Reconstructions of the Hernando de Soto expedition's route identify Parkin as Casqui, the seat of a paramount chief the expedition met in 1541 (Hudson et al. 1990; Mitchem et al. 2017; Phyllis A. Morse 1990:125–129). Because of its size, fortification, platform mound, and ethnohistoric identification, Parkin is routinely interpreted as the center of a Mississippian polity, and extending that interpretation from the town to the phase makes the further claim that the sites assigned to the Parkin phase belonged to a bounded, hierarchically organized society headed by the Parkin site.

We analyze the St. Francis basin phases to ask what the evidence supports, in a sequence of questions. Why do culture historians see phases in these data? If the phases are more than geographic clumps of assemblages structured by the landscape of rivers and waterways, is there evidence of social boundaries between them? That evidence is what would support the culture historians' claim. And what would it take, on a record like this one, to detect such boundaries? The analyses that answer these questions were built on data from Phillips, Ford, and Griffin (1951), compiled by Lipo (2001), using one spatial model, one neutral copying model as the standard of comparison, and two models built to detect bounded units. In this study, a bounded learning community means a population in which copying was more common internally and less common across its edge

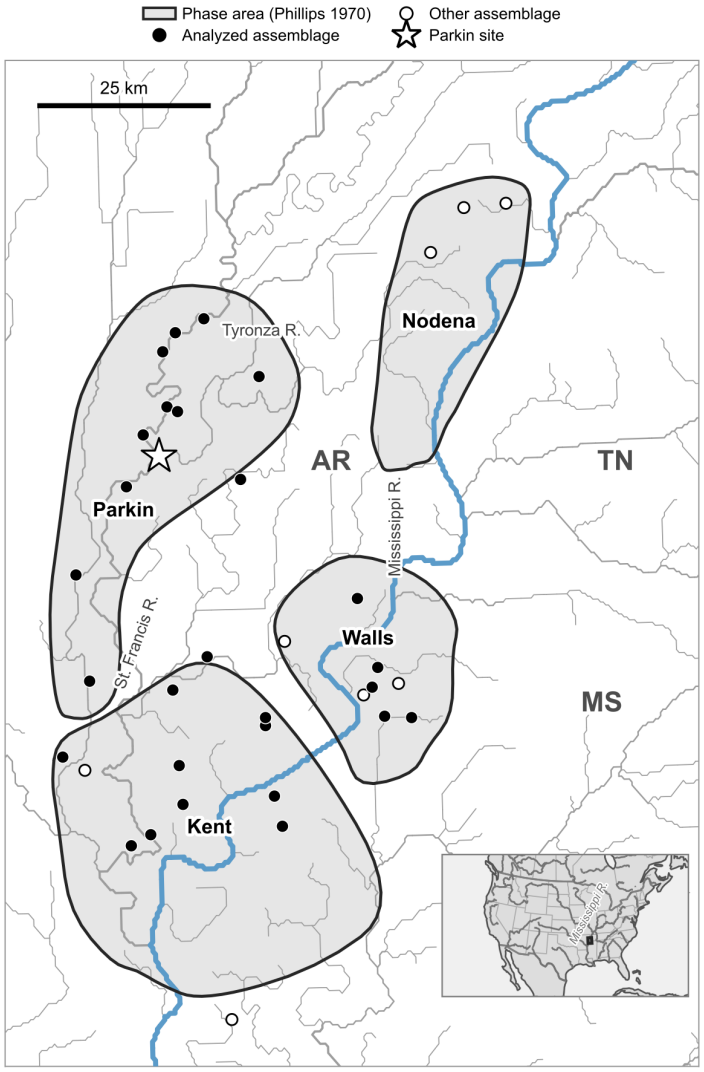


Figure 1. The study area and the late pre-contact phases as Phillips (1970) defined them. The Nodena, Parkin, Walls, and Kent phase areas are redrawn from Lipo (2001:Figure 2.6), after Phillips (1970), and placed on the map using the survey-quadrangle frame from that figure. Filled circles are the 28 decorated-ceramic assemblages analyzed here, open circles are the eight other assemblages of the 38 used in this study that fall inside the frame, and the Parkin site is starred. Phase areas are drawn over the HydroRIVERS river network (Lehner and Grill 2013), with line width scaled by flow order. The mapped areas illustrate the premise under test: that each contained a distinct community whose members learned mainly from one another. The mapped area lies in Arkansas, Tennessee, and Mississippi. Missouri lies just north of the frame. Inset: location in North America.

than distance alone would predict. Evidence for such a community requires more than geographic clustering. These analyses concern the organization of ceramic learning, and they cannot by themselves establish whether a political community existed or where its borders lay.

Why phases look like groups of people

Cultural transmission theory links patterns in pottery decoration to how people learned. Potters learned from others, so the designs they made preserve traces of those connections. In a neutral model, no decorative variant has an advantage in being copied. New variants arise, and their proportions change by chance as a finite population of potters copies existing designs, a process called neutral drift (Bentley et al. 2004; Dunnell 1978; Eerkens and Lipo 2005, 2007; Fogarty et al. 2024; Neiman 1995). Four expectations follow, and they organize what we measure.

The effects of copying generally fall off with distance. People learn from those they live near and see often, so a potter is more likely to copy a neighbor than someone further away. Under neutral drift, this tendency makes nearby assemblages more alike than distant ones, with similarity declining gradually across the landscape. Population geneticists call the pattern isolation by distance (Wright 1943), and it has long been recognized in stylistic frequencies (Dunnell 1970; Lipo et al. 1997; Neiman 1995; Shennan et al. 2015). It arises wherever interaction is mainly local, and it produces between-place differences on its own because small or poorly connected places drift toward idiosyncratic combinations of stylistic features (Lipo et al. 2021).

Settlement is clumped. People did not live evenly across the St. Francis basin. Settlements concentrate on particular reaches of the basin's rivers, with less-occupied stretches between them. So any gradual change in style is sampled unevenly, densely in some places and sparsely or not at all in others.

Gradual variation sampled at clumps looks like distinct units. Seriation and ordination sort assemblages by resemblance, and the culture historians who drew the phases also sorted them by proximity. When styles change gradually across clustered settlements, both criteria bring the assemblages of each settlement cluster together and place boundaries in the gaps. Units produced this way look distinct even when the copying process has no edge (Lipo et al. 1997, 2021). This is the difference between typological and population thinking (Dunnell 1980; Mayr 1959). Interpreted typologically, a cluster of similar assemblages is a unit with members and an edge. Interpreted as a population, the same cluster is a sampled portion of a larger learning network. Figure 2 shows the consequence for a record like this one: the same sampled values are consistent with a gradient and with a step at the gap, and the two histories differ only where no assemblage was collected.

Identifying a bounded learning community requires more than clustering. Restricting copying across an edge increases differences between communities beyond

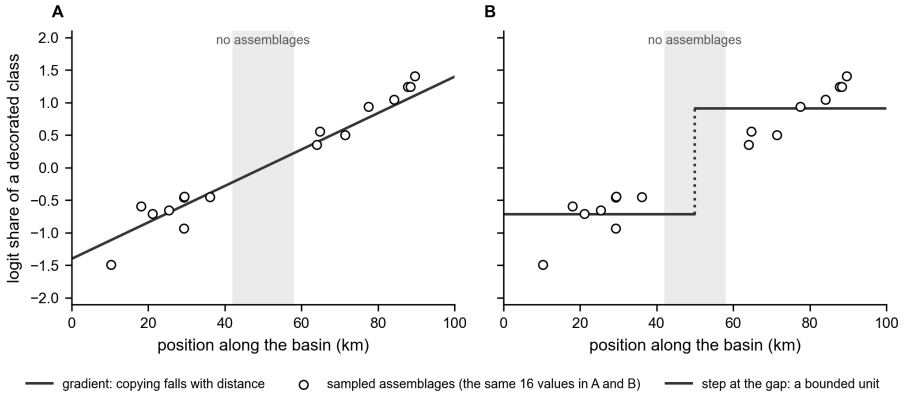


Figure 2. Two histories, one record showing an illustrative schematic, not data. Sixteen synthetic assemblages sit in two clusters along a transect with a stretch between them from which nothing was collected. The same sixteen values are consistent with a class's share changing gradually across the basin (A) and with a step at the gap, as a bounded unit would produce (B). The two histories differ only in the uncollected stretch. Whether a model can tell them apart on the real 28 sites is a measured question, answered for two models in Figure 7; this figure states the problem and makes no claim beyond this sample.

what distance alone produces, and it should do so across every independently transmitted aspect of the pottery, because a social boundary acts on the learners rather than on one design (Crema et al. 2016; Henrich 2001; Henrich and Boyd 1998). Whether such a restriction is detectable depends on its strength and duration, on where the sites happen to lie, and on the resolution of the record. Our standard for evidence of a social unit is accordingly that signatures recorded independently of one another converge. A single signature can be produced by several processes, and on this record two of the candidate signatures cannot serve at all. Deterministic seriation is built from the same frequencies that drew the phases, so its structure cannot test them, and tests of departure from neutral copying are distorted by the time-averaging of surface collections at this resolution (Madsen 2012; Premo 2014). What remains is the spatial signature of the decorated classes, interpreted against what distance-dependent copying produces, and the one trait system in these collections that was recorded separately from the decorated types.

Materials and methods

Data

The ceramic data are decorated-class counts from the Lower Mississippi Valley Survey collections of Phillips, Ford, and Griffin (1951), as compiled by Lipo (2001). We focus on decorated classes as candidates for neutral stylistic variation,

designs whose frequencies may change through copying and chance without affecting how a pot performs (Dunnell 1978; Neiman 1995). Variation in plain wares includes differences in paste and temper that can reflect firing conditions and vessel function (Feathers 2006; Feathers and Peacock 2008), and combining plain and decorated wares would also introduce variation in how much pottery was decorated. Ten decorated classes remain after pooling. Following Lipo (2001), we pool sherds from Barton Incised, Kent Incised, and Mound Place Incised, and we pool the painted types Old Town Red, Nodena Red and White, and Avenue Polychrome. Each class is a culture-history type, a unit of measurement built by analysts, not a category assumed to have existed in the past.

An assemblage takes the phase of the area it lies in, using the phase areas of Phillips (1970) as Lipo (2001:Figure 2.6) mapped them (Figure 1). Two assemblages fall outside every area and are assigned to the nearest, Cramor Place and Holden Lake, each 1.0 km outside. We include only assemblages with at least 75 decorated sherds, leaving the Nodena phase with a single assemblage, which we exclude. These rules leave 28 assemblages with 14,101 decorated sherds (median 251 per assemblage; range 79 to 1,709) across three phases: Parkin with 12 assemblages, Kent with 11, and Walls with 5. Of the 28, 18 are the 1940 survey collections alone, nine combine the 1940 collection with controlled surface collections Lipo made at the same site in 1996 and 1997, and one, Holden Lake, is Lipo's alone (Lipo 2001). We located each assemblage using the Lower Mississippi Survey site files, and we drew settlement attributes from a regional database (Lipo and Dunnell 2007). Mainfort (2003a) compiled a separate database from largely the same survey files with revised phase assignments. Because its counts overlap ours, we analyze it separately as a second case in the Supplemental Text.

These assemblages are time-averaged surface collections, built up over years of use and discard, and we treat them as the accumulated product of copying across a population rather than as degraded samples of a single moment (Dunnell 1995; Holdaway and Wandsnider 2008; Lipo et al. 2015; Neiman 1995; Perreault 2019; Schiffer 1972). Time-averaging damps differences between assemblages and distorts measures of diversity and neutrality in proportion to how long variants remained in use (Madsen 2012; Miller-Atkins and Premo 2018; Premo 2014) [CITE-CHECK: Premo 2014]. A neutral model also describes people copying one another, whereas an assemblage counts durable objects, so its expectations apply only approximately (Crema et al. 2024).

Radiocarbon determinations compiled by Mainfort (2001) match four of the 28 assemblages, with pooled calibrated medians of AD 1466 at Kent Place, 1483 at Parkin, 1513 at Neeleys Ferry, and 1597 at Clay Hill, and Hollywood has two calibrated one-sigma ranges reaching back to AD 1285 (Haley 2014; House 1993:23). On that evidence, the late pre-contact occupation spans roughly the fourteenth through the early seventeenth centuries, and the sequence records communities affected by European contact and the demographic collapse that followed (Lipo 2001; Morse and Morse 1996). The first axis of a correspondence analysis of the decorated frequencies tracks latitude (rank correlation +0.77;

latitude and longitude together account for 63 percent of its variance), and the two earliest dated assemblages share one end with the latest. The results show, however, that the deposits overlap temporally and accumulated over time. Thus, every analysis below treats the record as a time-averaged spatial sample, and the neutral simulations it is compared against are built to the same condition.

The analyses, by question

Table 1 lists the analyses in the order of the questions they answer, with what each measures and what it is compared against. Three kinds of numbers appear in the Results. Posterior quantities state what the record implies about a quantity once the uncertainty in its counts is carried through. Neutral comparisons place an observed statistic against the distribution of the same statistic over records simulated under neutral copying. Descriptive statistics summarize the record or a set of divisions and carry no inferential claim. No inference in this paper rests on a significance test, a p-value, a Bayes factor, or a binary threshold.

Table 1. The analyses, by the question each answers. Findings files are in the public repository (Data and Code Availability).

Question	Analysis	What it measures	Compared against	Kind of number
Why do culture historians see phases?	Clustering of the 28 assemblages on coordinates, on composition, and on both	Agreement with Phillips's phase assignments (adjusted Rand index over the 26 assemblages inside a phase area)	Divisions of the same sizes placed at random	Descriptive
	Spatial field model (A0)	Share of between-site variation in composition that lies in a smooth spatial surface; contrast between sites as a function of distance	Prior sensitivity under three intercept priors	Posterior

Question	Analysis	What it measures	Compared against	Kind of number
	Settlement density along the reconstructed phase boundaries	Late-period sites per 100 km² along the boundaries, from 53 sites independent of the assemblages	Boundaries of 300 divisions of the same sizes, random and compact	Descriptive
Is there evidence of social boundaries beyond geography?	Cultural F_ST between the two spatial clusters the site layout supports	Share of decorated diversity lying between the clusters	Neutral copying model at two operating points, 30 cells	Neutral comparison
	Per-class between-site variation	Moment F of each common class; sites present for each rare class	Neutral copying with and without regional drift of the innovation target; planted markers	Neutral comparison
	Composition by collection	The same statistics on the 1940 collections alone; per-site contrasts between the 1940 and 1996 to 1997 collections	Posterior under a Dirich-let(1/2) prior per collection	Posterior

Question	Analysis	What it measures	Compared against	Kind of number
	Rim attributes on 19 sites	Moment F of interior beveling, exterior notching, lip form, and Bell paste; orientation with latitude and longitude; pair association with decoration after distance	The same neutral model on the 19 sites; the common decorated classes on the same sites	Neutral comparison and posterior
What would it take to detect boundaries?	Latent-unit model (A1) and barrier model (A3) on their own generators	Recovery of a planted between-unit scale or barrier	Planted truth; prior median	Posterior
	Planted markers in the neutral model	Planted records with a class outside the neutral band	Neutral cells' own count outside their band	Neutral comparison

The spatial field model

Our first analysis describes how pottery composition varies with geography without first dividing the sites into groups. We fit a Bayesian spatial field model, A0, to the counts of the ten decorated classes in each of the 28 assemblages. In the model, each assemblage is a multinomial sample, the distribution that describes sherds drawn at random from a site where the ten classes occur in fixed but unknown proportions. We do not estimate those proportions independently for each site. Each is built from two parts: a baseline for the class, which is the same at every site and reflects how common the class is across the basin, and a site-specific adjustment. Addition occurs on the log-odds scale, which is the logarithm of a class's proportion divided by the reference class's proportion, and the softmax function converts the ten log-odds into positive proportions that sum to one. One class serves as the reference, with its log-odds fixed at zero, and the model estimates the other nine classes relative to it.

Geography enters through the site-specific adjustments. For each of the nine non-reference classes, the adjustments across the 28 sites are modeled as a Gaussian process, a smooth surface over the map in which nearby sites take similar values and distant sites take nearly independent values. An exponential covariance sets how quickly that similarity declines. The correlation between two sites falls as exp(-d/rho), where d is the straight-line distance between them, and rho is the range, a distance in kilometers that governs how far the resemblance extends. A second parameter, the field scale sigma, sets how large the geographic adjustments can be. Each site also receives an adjustment with no spatial pattern, independent from site to site, whose scale tau sets how much a site can differ from its neighbors for reasons unrelated to location. All nine class fields share one sigma, one rho, and one tau. Comparing sigma with tau is the purpose of the model. A record organized by location places most of its between-site variation in the surface, and a record in which each site has its own character regardless of its neighbors places it in the site-level term.

We gave sigma and tau the same half-Normal(0, 2) prior, so that neither source of variation is favored before the data are seen. For the range rho we used a prior proportional to 1/rho over 0.5 to 300 km, which gives equal prior probability to each multiplicative interval of distance within that window and so expresses no preference between a range of a few kilometers and one longer than the basin. On the class baselines we placed a Normal(0, 10) prior, which on the log-odds scale is wide enough to let a rare class be very rare. A narrower prior on the baselines can shift part of a rare class's rarity into the field's basin-wide level, so we refit the model under two alternatives, a Normal(0, 2) prior and a Normal prior centered on each class's log-odds in the pooled counts of the whole basin, and report all three. We wrote the model in Stan and fitted it by Hamiltonian Monte Carlo (Carpenter et al. 2017).

The data's limits shape what we report. All 28 sites lie within 79 km of one another, and at those distances a field with a large scale and a long range produces nearly the same contrasts between sites as a field with a smaller scale and a shorter range. These data constrain how different two sites are expected to be, not sigma and rho separately, and we report sigma and rho as not identified. Two quantities the data do identify are reported instead. The first is the contrast-based spatial share. A class's field has a basin-wide average level that the data cannot separate from the class baseline, so we remove that level and take the variance of what remains, the part of the field that makes sites differ from one another. This share is the remaining variance as a fraction of the total variance plus the site-level variance tau squared. A share near one means nearly all between-site variation the model finds lies in the smooth surface, and a share near zero means sites differ in ways location does not describe. Second, we report the standard deviation of the contrast between two sites a stated distance apart, on the log-odds scale, for example, the median spacing between nearest neighbors or the largest distance between any two sites in the sample. It states how different, under the field alone, two sites that far apart are expected to be in a class's log-odds.

The share is interpreted as the fraction of between-site variation in composition that a smooth geographic surface accounts for. It does not by itself distinguish a smooth surface from social units aligned with geography. If units existed and their members lived near one another, each unit's shared composition would appear at 28 sampled sites as one region of a smooth surface, and the surface would account for it. A high share explains why phases look like groups, which is the first question, and leaves open whether anything beyond geography structures the record, which the models described under Detecting bounded units address.

The neutral copying model

To state what distance-dependent copying predicts, we use the finite-population network model of Lipo, DiNapoli, Madsen, and Hunt (2021) on the real site map. Each of the 28 sites has a finite population of learners who copy decorated classes mostly within their site and, otherwise, from other sites with a probability that falls with straight-line distance. New variants enter at a constant rate in proportion to each class's share of the regional pool. Each simulated assemblage accumulates over a span with a random start and a random length, independent of location and phase, sized by the simulator's measured decorrelation length, and is subsampled to the real assemblage's sherd count. Nowhere does the model restrict copying.

Four model parameters cannot be determined from 28 time-averaged assemblages. The number of learners at each site sets how fast chance alone changes class frequencies, since a small population drifts faster than a large one. Innovation is the probability that a learner's class in a given step is drawn from the regional pool rather than copied from another learner. Mixing is the probability that a copy comes from another site rather than from the learner's own, and interaction length is the distance over which the probability of copying from a particular other site declines. Many combinations of these four produce records with the same diversity, and the record alone cannot distinguish among them.

We do not make the simulated assemblages resemble the real ones. If we tuned the parameters until the simulations matched the sites, the comparisons that follow would test nothing, because the model would reproduce the record by construction. Instead, we screen a grid of combinations of kernel shape, interaction length, population size, innovation rate, and mixing rate, and we keep those whose simulated assemblages match three summaries of diversity within a single assemblage, each within 10 percent of the record's value. These are the mean Gini-Simpson diversity, the probability that two sherds drawn from the same assemblage belong to different classes; the mean number of classes present in an assemblage; and the Gini-Simpson diversity of the 28 pooled assemblages. These summaries describe assemblages one at a time and provide no information about how sites differ, which the later comparisons measure. From the matched combinations we retain two operating points, chosen by a rule fixed before the screen was run. The primary point uses a Gaussian kernel with a 6 km interaction length, 2,000 learners per site, an innovation rate of

0.002, and a mixing rate of 0.4. A secondary point uses an exponential kernel at 6 km, 120 learners, an innovation rate of 0.024, and the same mixing rate of 0.4. These two differ in kernel shape, population size, and innovation rate while matching the same diversity summaries, so a result that holds at both does not depend on those choices.

Some predictions depend on a parameter the diversity match leaves open, for example, the span of time over which an assemblage accumulates or the amplitude of regional drift, both described in the next section. Such a parameter is not set to a single value. It is swept over a grid, and the record's position relative to the model is reported at every point on that grid, so readers can see whether a conclusion holds across the range or only at one setting.

As used here, the simulator states what neutral copying predicts for this site distribution, and the record is compared against that prediction. For each statistic, we generate 50 simulated records at an operating point, compute the statistic on each, and report the median and the 10th-90th percentile band across the 50. We classify the record as inside the band, outside it, or unavailable, the last when too few of the simulated records yield a finite value of the statistic for a band to be formed. Only an observed value above the band is interpreted as more variation than neutral copying produces, because restricted copying, if it occurred, would add to the differences between sites, and a record that varies less than the model does not indicate it. Every such interpretation is conditional on the generator and its operating point. It states what neutral copying predicts under these parameter values, and a different generator or setting could predict differently, which is why the swept parameters are reported at every point.

Per-class variation, regional drift, and planted markers

Cultural F_ST (Bell et al. 2009; Wright 1951) measures the share of decorated diversity, by the Gini-Simpson index, that lies between groups rather than within them, in Nei's (1973) form. We report it between the two spatial clusters the site layout supports, with a neutral band of 30 cells (the two operating points crossed with sweeps of accumulation span, span tied to collection size, and regional drift). For each decorated class, we compute a site-weighted moment estimate of F: the between-site variance of the class's share, with binomial sampling variance removed, divided by its mean share times one minus that share. For the three common classes (Parkin Punctated, the pooled Barton, Kent, and Mound Place Incised, and the pooled painted types), the statistic is F. For the seven rare classes, it is the number of the 28 sites where the class is present, because F is poorly determined for a class present at few sites. Neutral copying predicts one F shared by every class.

Change over time is a second source of differences between sites that has nothing to do with boundaries. Assemblages that accumulated at different times sampled a regional pool of designs that had itself changed in the interval, and regional drift, the rise and fall of class frequencies across the whole region through copying

and chance, is the documented form of that change in these data. As described so far, the simulator holds the regional pool from which innovations are drawn fixed. To let that pool drift, after each copying step we redraw it from a Dirichlet distribution centered on its current composition. Each class's expected share is unchanged by the redraw, and the size of the random step matches the mean and variance of one generation of a Wright-Fisher population, the standard model of change by chance alone. Over the occupation window, the stretch of simulated time from which the assemblage spans are drawn, the pool changes by chance alone, and we set that window to $W = 8L_R$, eight times the number of simulation steps over which the regional composition decorrelates. We quantify drift by its accumulated effect. The amplitude F_W is the regional F accumulated over the window, measuring how far the pool at the end of the window has drifted from the pool at its start, on the same scale as the between-site F reported for the classes. It is 0 when the pool is fixed, and we run it at 0, 0.01, 0.03, 0.1, and 0.3, the last larger than any between-site F the record shows.

A comparison that finds no departure is informative only if it could have found one. To measure that, we plant a marker, a known change in one class in one of the two spatial clusters the site layout supports, and ask whether the comparison detects it. Planting multiplies the odds of the class by 1.5, 2, or 3 at every site in the cluster, applied to each site's composition before its sherds are drawn, so that a class with a share of 20 percent at a site becomes 27 percent at a multiplier of 1.5, 33 percent at 2, and 43 percent at 3. Everything else in a planted record is identical to the neutral record it was made from. For each marker and each drift amplitude, we count how many of the 50 planted records have the class's statistic outside the neutral band for that class and amplitude. We report that count beside the number of neutral records that fall outside their own band, which is near 10 of 50 by construction because the band spans the 10th to the 90th percentile. A marker the comparison can detect raises the count well above that baseline, and one it cannot detect leaves the count near it.

Composition by collection

Nine of the 28 rows combine the 1940 survey collection with Lipo's 1996 to 1997 collections at the same site. To ask whether merging collections produces the per-class pattern, we decompose each row into its parts and recompute the per-class statistics on the 1940 material alone, matched on the 27 sites whose 1940 part has at least 75 sherds. At each mixed site we report the log odds ratio of each class between the two collections, with its 90 percent interval under independent multinomial sampling of each part and a Dirichlet(1/2) prior, and the medians under Dirichlet(0.1) and Dirichlet(1) as prior sensitivities.

Rim attributes

Mainfort (2003b, Table 1) tabulated rim attributes on 7,195 rims from 55 late-period assemblages, separately from the decorated types and by one observer. Nineteen of our 28 assemblages are among them, with 2,065 rims. We take three

attributes not tabulated as types in 1951, interior beveling, exterior notching, and lip form, and we treat paste (Bell Plain against Neeley's Ferry Plain) separately, because paste is a type-level distinction of the 1951 typology and is not independent of how the phases were drawn. For each attribute, we compute the same moment F on the 19 sites against the same neutral model, with all 28 populations transmitting and the 19 rim sites observed, beside the three common decorated classes on the same 19 sites. We report each attribute's rank correlation with latitude and longitude and its pairwise differences with distance, as posterior summaries under a Beta(1/2, 1/2) prior on each site's frequency (4,000 draws, Beta(1, 1) as sensitivity), and the rank correlation between rim and decoration differences across the 171 pairs of sites after linear adjustment on ranks for distance. Phillips (1970) did not define the phases on decorated types alone, so we describe the rims as a second trait system recorded by a second observer and do not call them independent of the phase definitions.

Detecting bounded units

The spatial field model describes composition as a smooth surface. Two further models ask whether the record is better described by that surface plus discrete units, each unit a set of sites that share a shift in composition beyond what their locations predict. Both keep everything in A0 and add one feature, so that A0 is recovered exactly when the added feature is set to zero. The latent-unit model, A1, assigns each site to one of three units, the number of phases in the basin, without being told which site belongs to which. Each unit shifts the log-odds of every class at its member sites, and the three units' shifts sum to zero within each class, so the units move composition relative to one another without changing the basin-wide baseline. A single scale, σ_g, governs the size of those shifts and has a half-Normal(0, 2) prior. Membership is unknown. A Dirichlet(1) prior on the frequencies of the three units expresses no preference among divisions, and the model sums over every possible assignment of sites to units rather than committing to one, so the posterior for σ_g reflects every way of dividing the sites into three units. A3, the barrier model, fixes the division in advance from geography alone, by drawing an edge through the site map, and asks whether sites on the same side of the edge share a shift in composition beyond what the smooth surface predicts. Its barrier parameter ω is the standard deviation, in log-odds, of that shared shift, with a half-Normal(0, 2) prior, and $\omega = 0$ is A0 exactly. We placed the edge in two ways from the site coordinates alone: the split that a threshold on edge lengths between neighboring sites produces, which separates the Walls area from the rest, and a straight line across the basin.

Before either model could be fitted to the record, it had to pass a test on data whose answer is known. We simulated records from each model's own generator at the real 28 site locations, with the planted value set at σ_g of 0.25, 0.5, 1, and 2 for A1 and at ω of 0, 0.25, 0.5, and 1 for A3, and fitted the model to each simulated record. For A3, we repeated the fits with the field's range left free and with it pinned at 6 and 12 km, and both models were fitted under

both class-intercept priors. A model that recovers the planted value returns a posterior centered near it that moves as the planted value moves. One whose posterior stays near the prior's median of 1.35, whatever the planted value, has learned nothing from the data. We report posterior medians against planted values, sampler diagnostics for every fit, and the share of fits that pass those diagnostics. A model that does not recover a planted value on records from its own generator cannot be interpreted on the record, where the truth is unknown, so we report no estimate from it.

Statistical conventions

We use three statistical conventions that apply throughout. First, a posterior quantity is reported as its median with a 90 percent equal-tailed interval, the range from the 5th to the 95th percentile of the posterior draws, unless stated otherwise. Given the data and the model, the quantity lies inside that range with probability 0.9.

Second, the sherds counted in an assemblage are a sample of what was deposited, so an assemblage's class proportions are uncertain, especially for small assemblages and rare classes. We carry that uncertainty into every statistic computed from proportions. Each assemblage's proportions have a Dirichlet posterior, the distribution that results from combining a multinomial likelihood for its counts with a Dirichlet(1/2) prior. That prior adds half a sherd to every class and keeps a class never observed at a site from being treated as impossible there. Each statistic that depends on proportions, such as F_{ST} or a per-class F, is recomputed on repeated draws from those posteriors, and its reported median and interval summarize that set of values. This interval reflects sampling uncertainty in the counts, not only spread between sites.

Third, every model fitted in Stan is checked for whether the sampler explored the posterior before any of its output is used. For every fit, we report five diagnostics. R-hat compares chains started from different points and is near 1 when they agree. The bulk and tail effective sample sizes give the number of independent draws the correlated chains provide at the center and extremes of the posterior. A divergent transition is a step at which the sampler's numerical integration failed, and each one signals a region of the posterior the sampler cannot traverse. Tree depth saturation is how often the sampler reached its configured step limit. The energy Bayesian fraction of missing information detects samplers that move too slowly between high- and low-density regions. A fit passes when R-hat is at most 1.01, both effective sample sizes are at least 400, no transition diverged, and the energy fraction is at least 0.3. We report tree-depth saturation but do not fail a fit, and for the latent-unit model we also require that the co-membership probabilities agree across chains. A fit that fails is reported as failing, and its posterior is not summarized, because draws from a sampler that has not explored the posterior do not represent it.

The Supplemental Text includes prior and posterior predictive checks, which

compare the counts a model generates before and after seeing the data with the observed counts; prior sensitivity; and recovery of planted values under each prior. Every signed quantity is stated with its direction in words, so readers can check the sign against the sentence that carries it.

Results

Why did culture historians see phases in these data?

The phases are recoverable from the site map's spatial distribution alone. Ward linkage on the 28 assemblages' coordinates, scored on the 26 assemblages that lie inside a mapped phase area, agrees with Phillips's assignments at an adjusted Rand index of 0.740, and the two assemblages it places with another phase, Big Eddy and Castile Landing, are the two in the narrow southern extension of the Parkin area, which it groups with Kent (Figure 3A). Divisions with the phases' own sizes placed at random agree at a median of 0.485, matching what group sizes alone produce. Composition alone, under a chi-square transform that needs no treatment of zero counts, agrees at 0.358 and misplaces six of the 26. Adding composition to the coordinates does not raise the agreement above the map alone under Ward linkage, and a k-means clustering on the coordinates reaches 0.480 and rises to 0.740 only when composition enters at a quarter of the coordinates' weight (Supplemental Text). A scheme divided by the map alone would be recovered and would not improve as the pottery data were included, and that is what these assemblages show. This result describes how the scheme was drawn by workers who sorted these collections by proximity and resemblance, and says nothing yet about how its units behaved.

Composition itself is organized by location. In the spatial field model, the contrast-based share of between-site variation that lies in the smooth surface has a posterior median of 0.93 with a 90 percent interval from 0.78 to 1.00 under the Normal(0, 10) intercept prior, and 0.93 (0.79 to 1.00) and 0.94 (0.78 to 1.00) under the two alternative priors (Figure 4). Between two sites, the contrast grows with their distance apart, from a standard deviation of 0.89 logits (0.70 to 1.10) at the median nearest-neighbor spacing of 4.3 km to 2.37 (1.91 to 2.85) at the median pair distance of 36.6 km and 3.12 (2.51 to 3.79) at the largest pair distance of 79.1 km. Scale and range are not separately identified at 28 sites and are not reported as estimates. All three fits pass every diagnostic (largest R-hat 1.006, smallest bulk effective sample size 608, no divergent transitions, no iterations at maximum treedepth, smallest E-BFMI 0.67). Phillips's areas also differ in pooled composition. Between the Parkin and Kent areas, the pooled logit share of Parkin Punctated is 0.47 higher in Parkin, and the shares of the pooled Barton, Kent, and Mound Place Incised and of the painted types are 0.12 and 0.90 lower.

The phase boundaries run through the least-occupied ground. We reconstructed boundaries by assigning each point on the map to its nearest analyzed assemblage, merging territories by phase, and measuring the density of late pre-contact

settlement along them using the 53 sites in the regional compilation that lie more than a kilometer from any analyzed assemblage. At a 5 km bandwidth, the density along the phase boundaries is 0.56 sites per 100 km² against 1.01 at the assemblages themselves, and 94 percent of 300 divisions with the phases' sizes around random centers and 100 percent of 300 compact divisions run through denser settlement than the phase boundaries do (Figure 3B). Results at 3 and 8 km are similar (Supplemental Text). Settlement clumps follow the larger rivers, and the boundaries lie in the stretches between them.

Taken together, these three results answer the first question. Culture historians saw phases in these data because pottery composition is organized by location, settlement clumps along the rivers, and sorting assemblages by proximity and resemblance places boundaries in the gaps. None of this restricts who copied from whom, and none of it rules anyone out.

Is there evidence of social boundaries beyond geography?

The difference between the two spatial clusters the site layout supports lies within what neutral copying produces. The between-cluster cultural F_ST is 0.0062, about six tenths of one percent of the decorated diversity, and it lies inside the neutral band in 17 of the 30 cells at the two operating points, including every cell at the primary point without regional drift. Cells above the band have the longest accumulation spans and the smallest drift amplitudes, which narrow the band from below (Supplemental Text). At the scale of the two clusters, distance-dependent copying on this map accounts for the record.

The decorated classes do not all behave the same way. Table 2 gives the per-class statistics of the record beside the neutral bands. All three common classes vary between sites far more than neutral copying alone would predict. Their site-weighted moment F values are 0.165 for Parkin Punctated, 0.117 for the pooled Barton, Kent, and Mound Place Incised, and 0.199 for the painted types, against neutral medians of 0.007 to 0.018 at every accumulation setting and both operating points. The seven rare classes, by contrast, are present at about as many sites as neutral copying predicts. At the primary operating point without drift, five of the seven lie inside the neutral band, Fortune Noded lies below it and Hull Engraved above it (Table 2), and the exceptions change with the cell; the rare classes' site counts are a weak discriminator, and the Supplemental Text reports them cell by cell. No setting of the simulator's unidentified parameters covers both the common classes' F and the rare classes' presence. Neutral copying predicts one F for every class, and the record does not show it.

Table 2. Per-class between-site variation of the record against the neutral copying model at the primary operating point without regional drift ($F_W = 0$; $W = 8L_R$; 50 records). For the three common classes, the statistic is the site-weighted moment F with binomial sampling variance removed; for the rare classes, it is the number of the 28 sites at which the class is present. Bands are the 10th to 90th percentiles over records. The two 27-site columns recompute

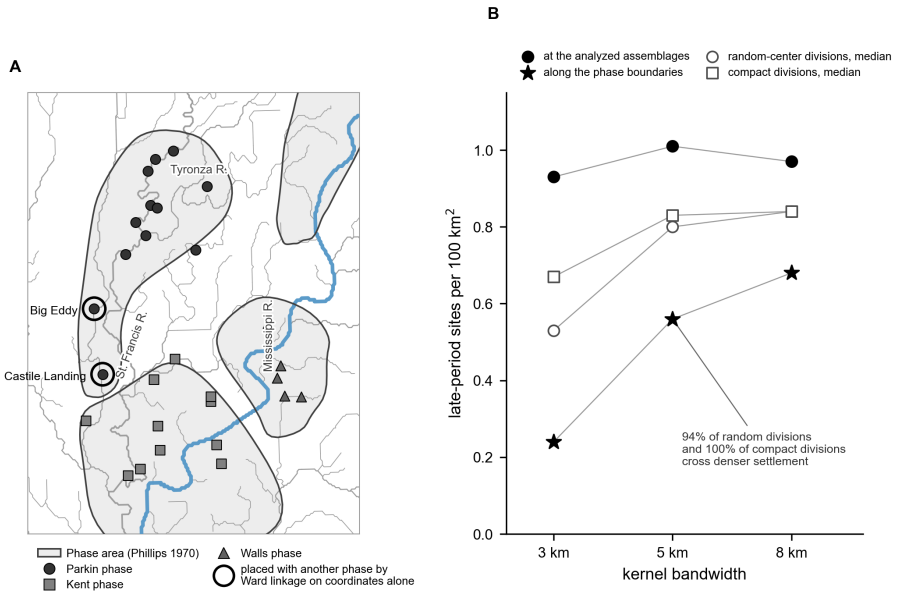


Figure 3. What recovers the phases. (A) The 28 assemblages colored by Phillips's phase, with the two assemblages that Ward linkage on coordinates alone places with another phase marked (Big Eddy and Castile Landing, placed with Kent). Agreement with the phase assignments across the 26 assemblages within a phase area is an adjusted Rand index of 0.740 for coordinates alone, compared with 0.485 for divisions of the same sizes placed at random and 0.358 for decorated-class composition alone. (B) Late-period settlement density, from the 53 compiled sites more than 1 km from any analyzed assemblage, at the analyzed assemblages, along the reconstructed phase boundaries, and (medians) along the boundaries of 300 divisions with the phases' sizes around random centers and of 300 compact divisions, at kernel bandwidths of 3, 5, and 8 km. At 5 km, 94 percent of the random and 100 percent of the compact divisions cross denser settlement than the phase boundaries do. Descriptive statistics, drawn from the committed summary table.

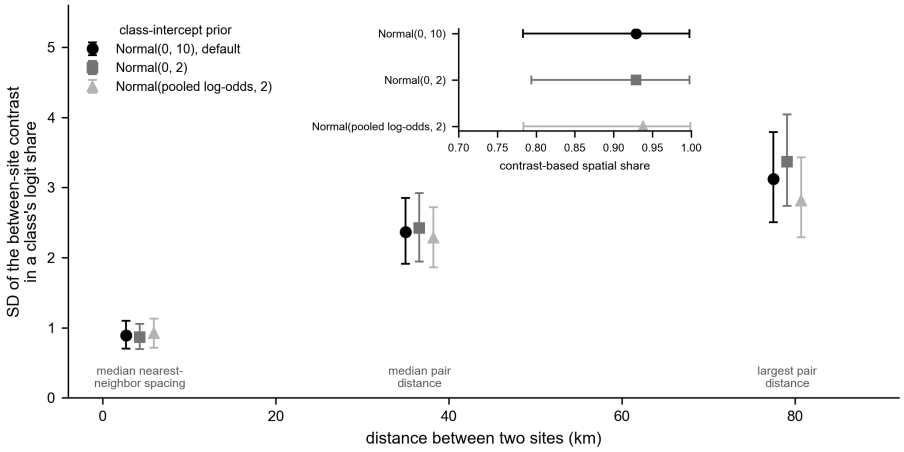


Figure 4. The spatial field. Posterior of the standard deviation of the between-site contrast in the logit share of a class as a function of distance, under the three intercept priors (Normal(0, 10), the default; Normal(0, 2); Normal centered on the pooled log-odds), with the median nearest-neighbor spacing (4.3 km), the median pair distance (36.6 km), and the largest pair distance (79.1 km) marked. Inset: contrast-based spatial share under each prior, with the median and 90 percent interval. Drawn from the posterior summaries of the three fits; the field's scale and range are not separately identified and are not shown.

the statistic on the 27 sites whose 1940 part has at least 75 sherds, first on the current counts and then on the 1940 survey collections alone; rare-class site counts in the 27-site columns are for the 1940 material. A status of "above" is the only outcome interpreted as more variation than neutral copying produces.

Class	Record (28 sites)	Record, current counts, 27 matched sites	Record, 1940 material, 27 matched sites	Neutral median (band)	Status
Parkin Punctated	F 0.165	0.145	0.151	0.013 (0.006 to 0.025)	above
Barton, Kent, and Mound Place Incised	F 0.117	0.076	0.087	0.013 (0.006 to 0.025)	above

Class	Record (28 sites)	Record, current counts, 27 matched sites	Record, 1940 material, 27 matched sites	Neutral median (band)	Status
Painted types	F 0.199	0.195	0.195	0.014 (0.007 to 0.021)	above
Fortune Noded	15 sites	not computed	12 of 27	19 (16 to 23)	below
Ranch Incised	22 sites	not computed	20 of 27	23 (19 to 25)	inside
Walls Engraved	14 sites	not computed	12 of 27	18 (14 to 21)	inside
Wallace Incised	7 sites	not computed	0 of 27	8 (5 to 14)	inside
Rhodes Incised	15 sites	not computed	11 of 27	11 (7 to 18)	inside
Vernon Paul Appliqué	10 sites	not computed	6 of 27	7 (3 to 12)	inside
Hull Engraved	8 sites	not computed	5 of 27	4 (1 to 6)	above

Regional drift over time does not remove the difference. Letting the regional innovation target drift over the occupation window raises the common classes' neutral F, but only to medians of 0.029 to 0.035 at the largest amplitude tried at the primary operating point, with 10th to 90th percentile bands reaching 0.052 to 0.111, and to medians of 0.033 to 0.041 at the secondary, and the record's 0.117 to 0.199 lies above every band (Figure 5). Drift also removes the rare classes from the regional target: at an amplitude of 0.01 the loss already runs from 1 to 44 of 50 records across the seven classes at the primary point, and at 0.1 Ranch Incised is lost in 39 of 50. All seven rare classes are covered by their bands together only at amplitudes of 0.01 and 0.03 at the primary point and at no tested amplitude at the secondary. No tested amplitude places the record inside the bands for the common classes, and the amplitudes that approach them remove the rare classes the record retains. Whether a neutral process with larger amplitudes, other occupation timings, or late-originating types could explain the difference was not measured and remains an open conjecture.

A moderate departure was expected. Doubling the odds of one common class in one of the two spatial clusters placed that class outside the neutral band in 23 to 45 of 50 planted records at amplitudes up to 0.1, and tripling it in 37 to 50 of 50, against the neutral cells' own 10 of 50 outside their band. A 1.5-fold change placed 8 to 26 of 50 outside, and moved the planted median outside

the band only for the pooled incised classes in the larger cluster at the two smallest amplitudes. Walls Engraved, the one rare class planted, never moved its occupancy median outside the band at any multiplier (Figure 5). These counts are conditional on the generator and the amplitude, and they show what size effect the neutral comparison detects in this sample, not what produced the record.

The departure is not an artifact of merged collections. On the 1940 survey material alone, matched on the 27 sites with at least 75 sherds in the 1940 part, the common classes' F values are 0.151, 0.087, and 0.195, versus 0.145, 0.076, and 0.195 for the current counts on the same 27 sites (Table 2). Collection source matters at the level of a site's two collections and for the presence of rare classes. At the six sites with large additions, the log odds ratio of Parkin Punctated between Lipo's collection and the 1940 collection runs from -1.03 at Cramor Place to +0.39 at Beck, with the 90 percent interval excluding zero at four of the six sites, at five of six for the pooled incised classes, and at one of six for the painted types. Wallace Incised is absent from every 1940 collection among the 28 and occurs only in Lipo's additions and at Holden Lake, so its presence pattern partly records which sites were revisited. A positive correlation between the pooled incised classes' share and collection size (Spearman +0.26 over the 28 rows) falls to zero in the 1940 material. Two correlations persist in the 1940 material. The share of Parkin Punctated rises with collection size (+0.42 over the 28 rows, +0.49 on the 1940 parts of the 27 matched sites) and the share of the painted types falls (-0.57 and -0.51). No neutral generator we ran reproduces them. Differences in site function may explain this. But these data cannot test that possibility.

Rim attributes recorded on 19 of the sites show their own structure. Interior beveling (F 0.105), exterior notching (0.069), and Bell paste (0.266) lie above every neutral band on the 19 sites, exceeding the band's upper limit by about 0.04, 0.04, and 0.21 or more in every cell (the smallest excess is 0.0395, for notching at the secondary operating point; Figure 6A). Lip form (0.031) exceeds the widest band by 0.0001, and a reconstruction of Mainfort's lip percentages to within one rim contains that edge, so the lip result is not resolved by the record. The three common decorated classes at the same 19 sites also lie above their bands (0.107, 0.044, and 0.208). However, the rims and decoration are oriented differently in space. Beveling and notching fall with latitude (posterior medians of the rank correlation -0.58, 90 percent interval -0.68 to -0.47, and -0.39, -0.53 to -0.25) and rise with longitude (+0.30 and +0.51), whereas Parkin Punctated rises with latitude (+0.65, 0.58 to 0.72) and the painted types fall with it strongly (-0.74, -0.78 to -0.70), with the pooled incised classes near zero (Figure 6B). After linear adjustment on ranks for distance, pairwise differences in each rim attribute and pairwise total-variation differences in decoration correlate weakly, with medians between -0.07 and +0.14 and intervals for beveling and notching against the current counts that include zero (Figure 6C). Rim attributes thus carry site-to-site structure beyond neutral copying that does not copy the decoration's structure within the same site. They provide a second line of

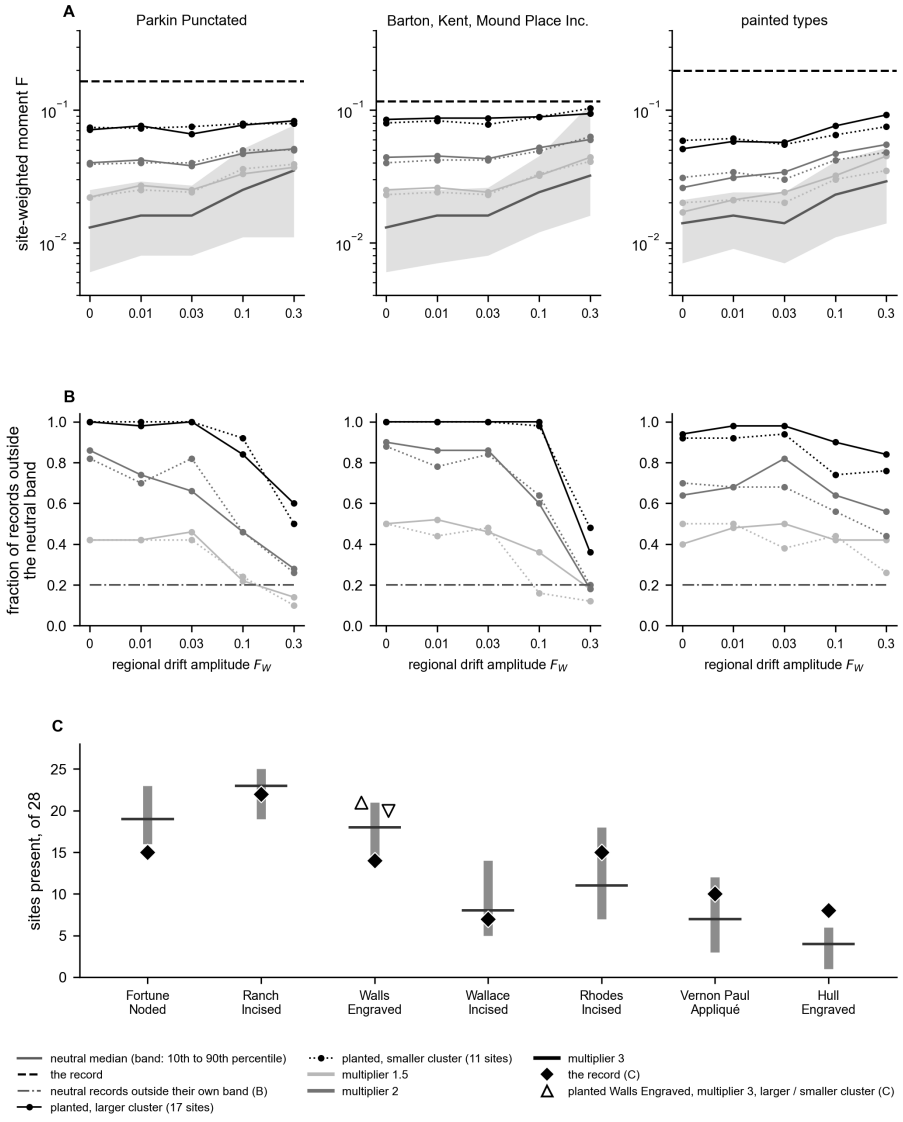


Figure 5. The common decorated classes against neutral copying with regional drift. (A) Site-weighted moment F of Parkin Punctated, the pooled Barton, Kent, and Mound Place Incised, and the painted types as the amplitude F_W of regional drift in the innovation target rises from 0 to 0.3 at the primary operating point ($W = 8L_R =$ 2,336 steps; 50 records per cell). Lines and bands are the neutral median and 10th-90th percentiles; the dashed line is the record; the colored lines are the planted medians for a marker in the larger and smaller spatial cluster at multipliers of 1.5 (solid), 2 (dashed), and 3 (dotted). (B) Fraction of planted records whose statistic falls outside the neutral band of the same class and amplitude, with the neutral records' own fraction outside their band (dash-dot). (C) Sites present for the seven rare classes at $F_W = 0$: neutral band (bar) and median (tick), the record (diamond), and the planted Walls Engraved occupancy at a multiplier of 3 in the larger and the smaller cluster (triangles). Drawn from the committed summary tables.

evidence for structure, recorded by one observer. How far the rims he tabulated overlap physically with the 1940 survey collections is unknown. They do not, however, bear on the phases as drawn, because rim attributes did not enter the phase definitions. Mainfort's (2003b:37) own account makes interior beveling spatial as well as temporal, so we cannot draw a chronological inference from it.

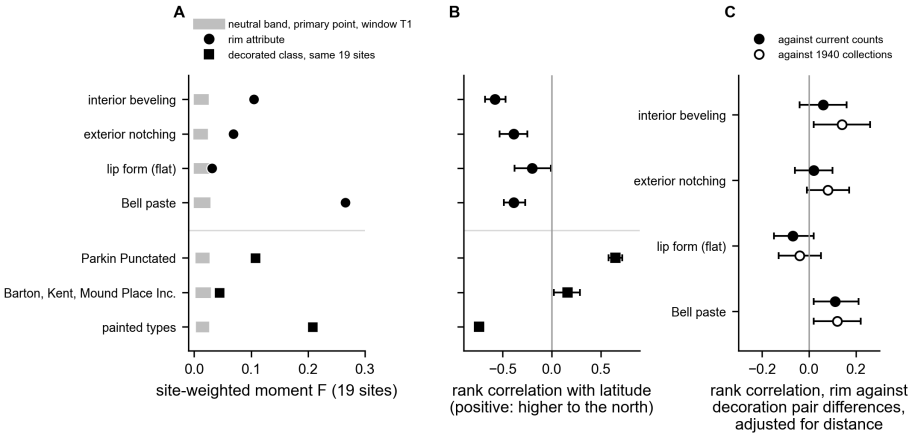


Figure 6. Rim attributes beside decoration on the 19 sites with rim tabulations. (A) Site-weighted moment F of interior beveling, exterior notching, lip form, and Bell paste, and of the three common decorated classes on the same 19 sites, against the neutral band at the primary operating point (window T1; 50 records). (B) Rank correlation of each attribute's and each class's site frequency with latitude, posterior median, and 90 percent interval under a Beta(1/2, 1/2) prior per site. Positive values indicate higher values to the north. (C) Rank correlation between rim and decoration pairwise differences across the 171 pairs of sites after linear adjustment for distance, against the current counts and against the 1940 collections. Positive means pairs that differ more in the attribute also differ more in decoration.

These results answer the second question in two parts. At the scale of the two spatial clusters, nothing in the decorated frequencies exceeds what distance-dependent copying produces. Below that scale, the three common classes vary between sites far more than neutral copying produces with or without drift over time; the rare classes do not, and the rim attributes vary beyond neutral copying along a different geography. The departure is real, but its cause is not clear. Three candidates are possible: differences in how collections were made or classified, which the collection decomposition demotes for this pattern but does not remove; directed change over time that is not drift; and spatial structure in the sense the culture historians meant. Only the third would indicate a social boundary. However, the patterns seen in the decoration alone do not separate this possibility from the other two.

What would it take to detect a boundary?

A model built to detect bounded units on this geography should recover them when we know they are there. We planted them, and neither model did. The latent-unit model A1 was fitted to records simulated from its own generator at the real 28 sites with the between-unit scale set to 0.5 logits, so the correct answer was known in advance. Its posterior medians for that scale came back at 0.27 to 0.48, below the planted value. On a record simulated with no units at all, where the correct answer is zero, the model returned 0.52 (0.05 to 1.39), an estimate as large as those from the records that contained units. An analyst given these posteriors alone could not tell a record with units from one without. Its sampler also failed to explore the posterior. Twelve of the 14 recovery fits failed the diagnostics: eleven on R-hat or effective sample size of the co-membership probabilities (the model's estimates of whether two sites belong to the same unit), and one on divergent transitions. Even the fit to the no-unit record failed on R-hat and bulk effective sample size (Figure 7A). Failed fits are reported as failed validation attempts, not as estimates. The failures are also not a matter of the intercept prior. Under the corrected Normal(0, 10) prior, the posteriors are essentially unchanged (0.58 against 0.52 on the no-unit record; 0.26 and 0.27 against 0.27 and 0.27 at a planted 0.5).

The barrier model A3 behaves differently and reaches the same end. Its sampler passes every check, but its posterior does not move with the planted barrier. With the range free and the Normal(0, 10) intercept prior, all 39 fits pass the diagnostics. At a planted barrier of zero, the model returned medians of 0.48 to 1.21 over six records, and at a planted barrier of 1.0 it returned 0.70 to 2.28 (Figure 7B). These two ranges overlap, so a posterior median from this model does not say whether the barrier in the record it was fitted to was absent or large. Both ranges center near the prior median of 1.35, where the posterior sits when the data say little about the parameter. Pinning the range at 6 or 12 km removes one parameter the data do not identify, and it moves the posteriors to 0.69 to 1.58 for a planted zero under the same prior and to 2.7 to 3.5 under the Normal(0, 2) prior, still without separating the planted values. All six prior-by-range combinations are tabulated in the Supplemental Text. A posterior that stays near the prior is the prior, not a detection. Such a model has returned what it was given rather than what the data contain.

The sites are the reason. These 28 assemblages sit in clusters separated by gaps of occupied but uncollected or sparsely occupied ground. A composition that changes gradually across the basin, sampled only at the clusters, is observed as a difference between the clusters, and a composition that steps at the edge of a bounded unit located at the same gap is observed as the same difference. Two histories leave one record (Figure 2). On these 28 sites, the two models that respect distance decay cannot tell them apart, and that is a measured statement about these two models and this sample, not a theorem about every model.

What the record would need is sampling where the two histories differ. Assem-

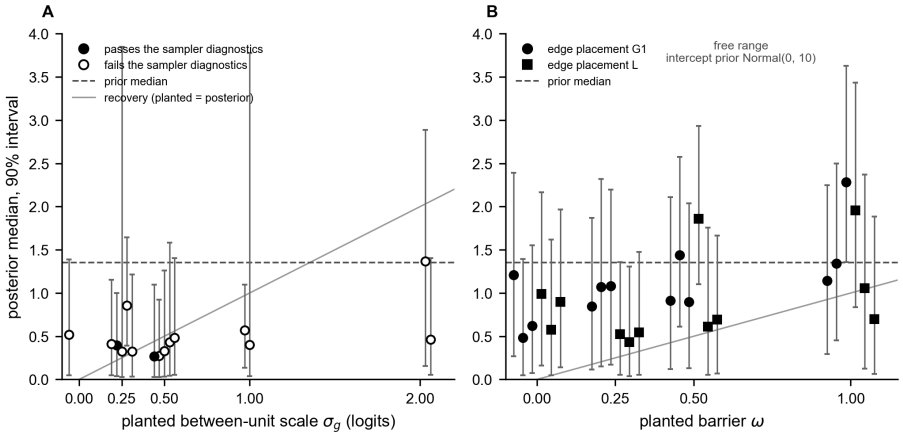


Figure 7. Recovery of planted structure on this site geometry. (A) Latent-unit model A1: posterior median and 90 percent interval of the between-unit scale σ_g against the planted value, on records simulated from the model's own generator at the 28 sites, with fits that fail the sampler diagnostics marked. (B) Barrier model A3 with the range free and the Normal(0, 10) class-intercept prior: posterior median and 90 percent interval of the barrier parameter omega against the planted value for the two edge placements, with the prior median (1.35) marked. The six prior-by-range combinations are tabulated in the Supplemental Text. Drawn from the committed recovery tables.

blages in the gaps would let a step be determined from a gradient, because a gradient predicts intermediate compositions there and a step does not. Dated or stratified contexts would let change along the sequence be observed, which these surface collections do not allow, and would show whether the common classes' extra variation is directed change over time at sites occupied at different times. Trait systems recorded independently of the phase definitions, of which the rim attributes are the one available example, would let the convergence criterion be applied. A social boundary acts on every independently transmitted trait, so a departure shared across trait systems along one geography is the expected signature, and the rims and decoration on these 19 sites do not share one. Finally, the neutral comparison registers a doubling of a common class's odds in one cluster in roughly half to nine tenths of planted records, so a departure of that size, if it were concentrated at one edge, is within what the method can detect on this sample. The record shows departures of that size and larger. What is missing is not resolution but the independent evidence that would attribute them.

Table 3. The three senses in which the phases could be real, and what the record shows for each.

Sense	Measured basis	Answer
A geographic pattern in pottery composition	Contrast-based spatial share 0.93 (0.78 to 1.00) under three intercept priors; coordinates alone recover the phases for 24 of 26 assemblages; boundaries in the least-occupied ground	Yes
Bounded units with edges	A1 and A3 do not recover planted structure on their own generators at these 28 sites; a step at a settlement gap and a gradient sampled at the gap leave the same record	Not resolvable on this sample, in either direction
Social units, under the convergence criterion	Seriation not independent of the frequencies; spatial signature not separable from distance; between-cluster F_ST 0.0062 inside the neutral band in 17 of 30 cells; rims and decoration structured along different geographies	Not demonstrated

Discussion

Three senses of "real"

The question of whether the phases are real covers three claims, and these data answer them differently. As a geographic pattern in pottery composition, the phases are real. Location organizes composition. Thus, the map alone recovers the scheme, and the boundaries lie where settlement is sparsest. This is the pattern the culture historians noticed, and their expectation that it would be there was correct. As bounded units with edges, the phases are not resolvable in either direction on this sample. The two models built to find edges cannot recover them on this geometry, so the record neither shows nor excludes a step at the gaps. As social units, which is what the culture historians meant and what the paper set out to test, the phases are not demonstrated. Our standard for that claim is that independent signatures converge, and on this record the signatures that could be assessed do not converge. Between-cluster differentiation

lies within distance-dependent copying, and the rim attributes that depart from neutral copying do so along a geography that is not the decoration's.

Phases are then best understood as what their makers said they were, slices of continuous variability drawn at the gaps in a clumped settlement pattern. McNutt (1996:250) wrote of the southern phases that "the distinctions that one tries so desperately to make between the phases involved are little more than comments on local elaborations—ceramic drift, if you will—of the basic pottery complexes possessed by peoples involved in this expansion," and McNutt, Franklin, and Henry (2012:247) describe the Walls phase as "a developmental lineage" spanning roughly two centuries, so that a phase's members need not even be contemporaneous. Our results give that interpretation a measured basis and also mark its limit. Mainfort's (2003a:180) observation that phase members lie "in the general vicinity of each other in ordination space" without forming clusters is what a geographic pattern sampled at clumps produces, and proximity on the ground accounts for proximity in the ordination.

What the record shows beyond geography

The common decorated classes vary between sites more than neutral copying produces at any setting we tried, with or without regional drift over time, and the rare classes do not. This is a measured departure from the neutral prediction of one F for every class, and it survives the removal of the later collections. We have deliberately left its three candidate causes unordered. Differences in collection and classification are less likely but not eliminated by the 1940-only comparison, because several field parties made the 1940 collections over a season. Directed change over time that is not drift remains possible, because sites occupied at different times during a span in which the common types rose and fell would differ in those types without any boundary between them, and the record does not have sufficient temporal information that would enable this option to be tested. Spatial structure in the sense the culture historians meant remains possible as well. Only the third option, of course, bears on the social interpretation of the phases, but decoration alone does not separate it from the other two. We note that the untested neutral form is a process with type origination, in which types enter the regional pool at different times, producing unequal per-class F under neutrality and potentially changing both the rare classes' occupancy and the per-class pattern.

Rim attributes complicate rather than resolve this. Beveling, notching, and paste depart from neutral copying on the same 19 sites by margins larger than the lip form's and comparable to the common decorated classes', and they decrease with latitude where Parkin Punctated increases with it. Two trait systems structured along different geographies is what independent transmission on one landscape produces, not what a social boundary acting on all learners produces. The rims are one observer's tabulation of 19 sites, with totals from 27 to 277, and whether they entered the phase definitions is unclear, so we interpret them as a second line of evidence that hints at structure rather than evidence against the phases.

Parkin

Parkin was the basin's largest town, with one of its tallest mounds, and its prominence is unquestioned. Decorated frequencies do not, however, establish a boundary in ceramic learning at the edge of the Parkin phase. The phase lies near the northern of the two spatial clusters the site layout supports, and the difference between those clusters reflects what distance-dependent copying produces. The mound-height distribution and regional evidence on hoe exchange are described in the Supplemental Text, and neither requires a restriction at the phase's edge. Other evidence for a Parkin polity addresses political organization and not the ceramic boundary. The region's own history offers a precedent for separating the two: Coosa, the best-documented Mississippian paramountcy, spanned two ceramic traditions and site clusters separable at the phase level, and political affiliation rather than shared decoration united them (Anderson 1996:250–251). If the polity the De Soto expedition met was at Parkin, the evidence for it lies in ceremony, fortification, and settlement size, and the decorated pottery does not resolve it.

Limits

Four limits are findings of this study, not caveats. Site geometry establishes the first. Because the assemblages sit in clusters separated by gaps, no model that respects distance decay can separate a step from a gradient in this sample, as the recovery tests for A1 and A3 measured. Chronology underpins the second. Four of 28 sites carry radiocarbon medians. The seriation order is built from these frequencies and tracks latitude, and no analysis here tests change along the sequence. Our neutral model is the basis for the third. Its operating points match within-site diversity and are not identified, and its fixed innovation target makes it a standard of comparison for spatial copying among contemporaneous sites, which is why the drifting target was added and why the per-class departure is reported at every amplitude rather than at one. The fourth is time-averaging, which damps differences between assemblages (Kidwell and Tomašových 2013; Perreault 2018; Tomašových and Kidwell 2009; Youngblood et al. 2023). Every simulated record carries accumulation spans of random start and length, so the reported bands include their effects under the modeled conditions.

What this approach contributes

Two requirements are needed for applications beyond this case. Interpreting a similarity pattern as evidence of a bounded social unit requires showing that the pattern exceeds what distance-dependent interaction on the actual settlement map produces, and the comparison must be made at matched diversity, because a neutral model run at the wrong diversity is easily credited with structure it cannot produce, just as a localized neutral process is easily mistaken for biased copying when diversity is taken at face value (Premo 2016; Premo and Scholnick 2011; Schneider et al. 2016). Before interpreting a model's boundary-detection results on a record, the model must recover a planted boundary on records

simulated at the same sites. The second requirement changed our conclusions. Had we fitted the latent-unit model to the record without the recovery test, we would have reported a posterior for the between-unit scale, and it would have told us nothing. Together with the convergence criterion, these requirements give a way to evaluate social interpretations of artifact variation within evolutionary archaeology (Dunnell 1995; Lipo et al. 2015; Neiman 1995), and they say what to collect next: assemblages in the gaps, dated or stratified contexts, and trait systems recorded independently of the units under test.

Conclusion

Phases organize the archaeological record of the Lower Mississippi Valley, and their use as social units makes a claim their definition does not: that similar pottery at nearby sites identifies a community and, at Parkin, a political territory. We asked why the phases look like groups of people and whether the St. Francis basin phases are more than geographic clumps. They look like groups because pottery composition is organized by location, settlement is clumped along the rivers, and sorting clumped assemblages by proximity and resemblance draws boundaries in the gaps. Coordinates alone recover Phillips's phases for all but two assemblages, a smooth spatial surface carries 93 percent of the between-site variation in composition, and the boundaries lie in the least-occupied ground. At the scale of the two spatial clusters, the decorated frequencies show nothing beyond what distance-dependent copying produces. Below that scale, three common classes vary between sites more than neutral copying produces with or without drift over time; the rare classes do not, and rim attributes recorded separately vary beyond neutral copying along a different geography. Neither model built to detect bounded units can distinguish them on this site geometry, because a gradient sampled at settlement gaps and a step at those gaps leave the same record.

The phases are real as a geographic pattern, unresolvable as bounded units on this sample, and not demonstrated as social units. Phase membership alone is weak evidence for a bounded learning community, and the inference from one phase to one polity is not warranted on its own. Resolving the question requires assemblages from the settlement gaps, dated or stratified contexts, and trait systems recorded independently of the units under test. Those are the data the culture historians' hypothesis has always needed.

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Data and Code Availability

All data and code needed to reproduce the analyses and figures reported here are openly available at https://github.com/clipo/phases. Upon

publication, we will release a versioned archive of the repository at https://doi.org/10.5281/zenodo.20796240. The repository includes decorated-ceramic class counts, site coordinates, rim attribute counts, and radiocarbon data; every analysis script with its computed outputs and figures; and instructions for reproducing all results. A manifest names the script that produces each result and figure, and every number in this paper is traceable to a findings file in the repository. The spatial field, latent-unit, and barrier models are Stan programs under `stan/spatial/`; the copying model is implemented in `src/mls_emergence/transmission/spatial.py`; the per-class, regional-drift, collection, and rim analyses are under `analyses/per_type/` and `analyses/rims/`. The copying-model and per-class results were produced under NumPy 2.5.4, and the remaining results under NumPy 2.4.6; because NumPy 2.5 changed the multinomial generator, the repository pins the version each result set requires. The phase tests are also packaged as a command-line tool, `phasecheck`, which applies them to any table of assemblage counts, coordinates, and phase assignments. The code is released under the MIT License, and the data under a Creative Commons Attribution 4.0 International license.

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